



DRISHTI

ENTERPRISE TECHNICAL SPECIFICATION

Drishti — UI/UX Design System & Guidelines

Defensive Cybersecurity Platform — Complete Architecture & Reference Manual

-  Graph-Powered Attack Paths
-  Live Watch Agent
-  Dollar Risk Quantifier
-  AI Remediation Guardrails
-  URL Trust Scoring

PLATFORM VERSION	RELEASE DATE	ARCHITECTURE TIER
v0.2 Enterprise Edition	August 2026	Full-Stack Multi-Tenant

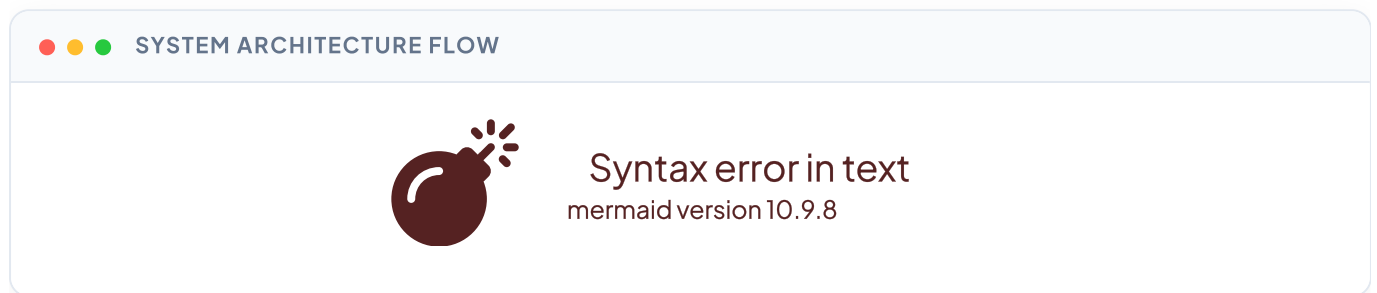
 **RESTRICTED // INTERNAL ARCHITECTURE & TECHNICAL SPECIFICATION // FOR AUTHORIZED USE ONLY**

Drishti — UX & UI Pattern Catalog

Reverse-engineered from the implemented frontend. Every screen, data contract, interaction pattern, visual language, and component layout — mapped from `web/src/`.

Last updated: 2026-08-17 — verified against source code.

1. Screen catalog (frontend routing)



Code-split chunks (3 public/app boundaries):

1. Landing/auth (separate lazy chunk) — never loads React Flow / Recharts
2. ProtectedApp (lazy-loaded) — only fetched after auth
3. Routes — each route is a lazy-loaded component

2. Visual language & theming

Light / dark mode

- CSS custom properties on `:root`; redefined under `@media (prefers-color-scheme: dark)`
- Three states: explicit `data-theme="dark"`, explicit `data-theme="light"`, system default (prefers-color-scheme)
- All colors use token references — never hard-coded values

Color system (token names)

Token	Purpose
<code>--bg-primary</code> / <code>--bg-secondary</code>	Page, card, elevated
<code>--text-primary</code> / <code>--text-secondary</code>	Headings, body
<code>--accent-primary</code>	Interactive elements
<code>--border</code>	Dividers
<code>--shadow</code>	Card elevation

Typography

- System font stack (not imported web fonts)
- Three sizes used: `text-2xl` (section), `text-xl` (sub-section), `text-sm` (table/compact)
- Weights: 400, 500, 600, 700

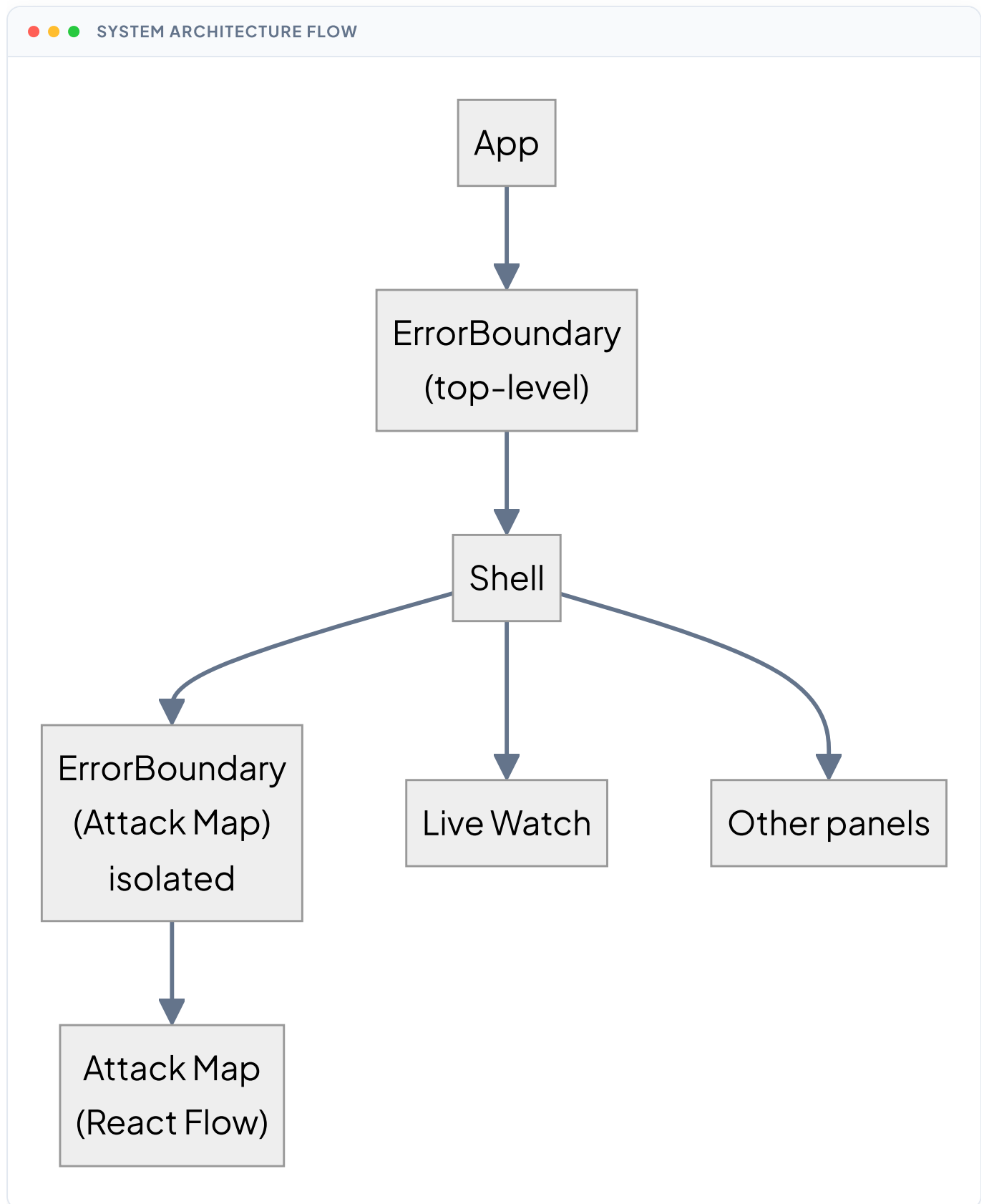
Motion

- **Framer Motion** (imported as `motion` / `AnimatePresence` / `LayoutGroup`)
- Shared layout animations between graph / panel / sub-panel
- Entrance: `initial={{ opacity: 0, y: 8 }}` → `animate={{ opacity: 1, y: 0 }}`
- Exit: `exit={{ opacity: 0 }}` with `transition={{ duration: 0.15 }}`
- Corner duration: `0.15–0.3s` (never long)
- List stagger: `staggerChildren` with `staggerDelay: 0.05` for card grids

Panels

- Left: **Graph panel** — wider (controls + subject panel)
- Right: **Scope panel** — narrower (context + status + filters)
- On the homepage (solo, no right panel): click a node → detail expands inline
- Sub-panel behavior: `// --- entry mode ---`, `// --- detail mode ---`, `// --- entity mode ---` — each is a conditionally rendered `<section>` swapping the right panel content
- Network detail is **never** collapsible — it always fills the panel

3. Error boundaries



- **Top-level boundary:** catches everything, replaces the app with a recovery message
- **Attack Map boundary:** isolated from the rest — if React Flow throws, other panels continue

- No per-panel boundaries on panels (a panel failure renders the panel area, not the whole app)

4. Motion catalog (by screen)

Attack Map (React Flow)

- Nodes fade in from center
- Selected node: handles show on demand + slight scale
- Hover: elevation change (shadow shift) rather than size change
- Edge highlights: `onTopPath` animated via React Flow's built-in `animated` prop
- Offline device overlay: colored dot opacity transition

Dashboard

- Stat cards: each is a `motion.div` in a `StaggerContainer`
- Zone summary: accordion-style expand (see Accordion)

Live Watch (force graph)

- Nodes repopulate gradually (`onLoad` callback in `useForceEffect`)
- Particle effect on threat nodes (red pulse — CSS animation)
- Demo attack injects trigger a re-render with label overlays

Remediation Console

- Remediation card: `LayoutGroup` for shared layout (card $\leftrightarrow$ detail)
- Expand/collapse: `AnimatePresence` for enter/exit
- Copy-to-clipboard: hover reveal of copy icon + click feedback
- Fix now: full-screen error view with retry/skip actions + details

5. Accordion (domain Accordion primitive)

State	Behavior
Closed	Title + risk badge + count
Closed + hover	Subtle background lift
Open	Content area mounts; flow animation
Open + click another	Previous closes (single-open), not toggle

Used by: Dashboard zone breakdown, Attack Paths summary, Network threats panel.

6. Keyboard & navigation

Shortcut	Screen	Action
Escape	All	Close modal/panel
Enter	Dashboard/Report	Scroll to section
Tab	All	Standard tab order
Ctrl/Cmd + K	Dashboard	Global search (planned)

Focus ring: visible, custom branded color (not browser default).

7. Hover and click guidelines

Pattern	Implementation
Table row hover	bg-primary highlight (subtle)
Button hover	bg-color shift + shadow elevation
Clickable card	cursor pointer + subtle shadow change on hover
Copy-to-clipboard	Hover reveals icon; click triggers toast; also shows a "Copied!" toast after click
Interactive edge	React Flow hover highlights connected nodes (debounced)

8. Copy patterns

Confirmation

i

NOTE

"Are you sure? This will [description]."

Error recovery

i

NOTE

"We couldn't [action]. [Specific reason]. Try [recovery step] or contact support."

Empty states

 NOTE

"No [items] yet. [CTA description]."

AI disclaimer (always shown)

 NOTE

"Outputs are AI-generated. Validate in your environment. This is advisory guidance, not a definitive fix."

Token issuance (agent)

 NOTE

"Your agent token has been generated. Copy it now — it will not be shown again."

9. Toasts

Type	Color	Icon	Use case
Success	Green (theme-token)	Check	Copy, generate, save
Warning	Amber (theme-token)	Triangle	Deep scan unavailable, auto-resolve
Error	Red (theme-token)	Cross	AI failure, network error
Info	Blue (theme-token)	Info	Recompute triggered

Duration: 4s auto-dismiss. Stack up to 3; older ones are replaced. Not interruptive for the system — the user dismisses explicitly if needed before timeout.

10. Form patterns

Pattern	Validation	Error display
Auth forms	Email format + password min 8 chars	Inline, beneath each field
URL analyze	URL format + reachable host	Inline beneath input
Deep scan	IP/CIDR + consent checkbox	Inline beneath each control
Settings toggle	Boolean switch	Immediate state update (no save button)
Asset edit	Title + optional fields	Inline, beneath each field

11. Loading states

Component	Loading state
Graph	Skeleton placeholder + spinner in node area
Dashboard	Skeleton cards (animated shimmer)
Tables	Row skeleton (4–5 rows)
AI remediation	Skeleton card (2–3 lines)
Deep scan	Full-screen modal with progress bar + nmap output streaming

12. Theme-aware patterns

Concern	Approach
Graphs	Colors adapt via token refs — no hard-coded node/edge colors
Severity colors	CSS custom properties or themed classes
Threat pulse	CSS animation (red) — works in both themes
Edge thickness	React Flow <code>style={{ strokeWidth }}</code> — theme-independent

13. Component inventory (key UI components)

Component	Location	Purpose
Shell	web/src/features/dashboard/components/Shell.tsx	Top-level nav + page layout
ErrorBoundary	web/src/components/ErrorBoundary.tsx	Catches render errors
AcmeLogo	web/src/components/acme-logo.tsx	SVG logo
StatusBar	Live Watch	Device count, online/offline
GraphCanvas	Attack Map	React Flow wrapper
ForceGraph	Live Watch	Force-directed layout
ThemeToggle	Settings	Light/dark mode switcher
ToastContainer	App level	Toast stack
TokenCopy	Remediation	Copy-to-clipboard
RecomputeButton	Shell	Trigger full recompute
ZoneBadge	Tables	Criticality/zone colored badge
RiskScoreGauge	Dashboards	Semi-circular gauge (risk score)

14. State management (client-side)

Concern	Library	Scope
Server state (API)	TanStack Query	All <code>api/</code> calls, cached by query key
Graph focus	Zustand (<code>graphStore.ts</code>)	Selected node, zoom, pan
Toasts	Zustand	Notification stack
Theme	CSS custom properties + context	Light/dark/system

15. Map type annotations (density reference)

Type	Example	Enforce level
page_id	<code> '/attack-map'</code>	allow — route string
field_id	<code> 'risk_score'</code>	allow — data key
token_id	<code> 'drishti_<base64>'</code>	allow — agent token
safe_html	<code> `description`</code>	<code> title`</code>
state_hash	<code> sha256(bytes).hex():16]</code>	allow — collision-resistant
url	<code> 'https://nvd.nist.gov/...'</code>	allow — fetched externally